

SOUGATA SAHA
IT-A2-037
ASSIGNMENT 2

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- a. Put a breakpoint in 1 st executable line of the innermost loop.

Ans:

```
(gdb) list
1      #include <stdio.h>
2
3      int main() {
4          int i, j, k;
5          for (i = 0; i < 10; i++)
6              for (j = 0; j < 200; j++)
7                  for (k = 0; k < 3000; k++) {
8                      int t1, t2;
9                      t1 = i;
10                     t2 = j;
(gdb) break 8
Breakpoint 1 at 0x400553: file d.c, line 8.
```

- b. If you run and continue ,how many times it is supposed to stop at breakpoint 1?

Ans:

```
[Inferior 1 (process 13748) exited with code 0347]
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el
(gdb) info breakpoints
Num      Type             Disp Enb Address                  What
1        breakpoint      keep y   0x0000000000400553 in main at d.c:8
          breakpoint already hit 6000001 times
          silent
          continue
(gdb)
```

c. How will you continue so that it stops at 1000 th iteration of innermost loop ?

Ans:

```
[Inferior 1 (process 3080) exited with code 0347]
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el
7_3.2.x86_64
(gdb) info breakpoints
Num      Type             Disp Enb Address                  What
1        breakpoint       keep y   0x0000000000400553 in main at d.c:8
          stop only if k==1000
          breakpoint already hit 2000 times
          silent
          continue
```

d. How you can condition your breakpoint , so that the loop stops at every 1000 th iteration of innermost loop ?

Ans:

```
(gdb) info breakpoints
Num      Type             Disp Enb Address                  What
1        breakpoint       keep y   0x0000000000400553 in main at d.c:8
          stop only if k%1000==0
          breakpoint already hit 6000 times
          silent
          continue
```

e. Put a breakpoint in the 1 st line of outermost loop.

Ans:

```
(gdb) info breakpoints
Num      Type           Disp Enb Address              What
1        breakpoint     keep y   0x0000000000400553 in main at d.c:8
          stop only if k%1000==0
          breakpoint already hit 6000 times
          continue
2        breakpoint     keep y   0x0000000000400541 in main at d.c:6
          breakpoint already hit 10 times
          continue
```

f. Disable breakpoint “1”

Ans:

```
[Inferior 1 (process 13453) exited with code 0347]
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el7_3.2.x
86_64
(gdb) info breakpoints
Num      Type           Disp Enb Address              What
1        breakpoint     keep n   0x0000000000400553 in main at d.c:8
          stop only if k%1000==0
          continue
2        breakpoint     keep y   0x0000000000400541 in main at d.c:6
          breakpoint already hit 10 times
          continue
(gdb)
```

g. Add a command to breakpoint 2 so that it prints the value of “i” at each hit.

Ans:

```
[Inferior 1 (process 14502) exited with code 0347]
(gdb) info breakpoints
Num      Type           Disp Enb Address              What
1        breakpoint     keep n   0x0000000000400553 in main at d.c:8
          stop only if k%1000==0
          continue
2        breakpoint     keep y   0x0000000000400541 in main at d.c:6
          breakpoint already hit 10 times
          print i
          continue
```

h. Delete breakpoint 2.

Ans:

```
[Inferior 1 (process 17299) exited with code 0347]
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el7_3.2.x
86_64
(gdb) info breakpoints
Num      Type             Disp Enb Address            What
1        breakpoint       keep n   0x0000000000400553 in main at d.c:8
          stop only if k%1000==0
          continue
```